

BROKILD · WINDOWS VST3 · KICK DRUM SYNTHESISER

/// KICKSTART

Every kick you need, from one small set of knobs. No samples.

MANUAL · BUILD 260926.1

BODY

PITCH

SWEEP

BEND

DECAY

CURVE

WAVE

SKIN

52.0 Hz

24.0 st

22 ms

420 ms

15 %

35 %

0 %

HIT

CLICK

TONE

60 %

70 %

ERA

GRIT

ROOM

0 %

0 %

TRANSIENT

ATTACK

SUSTAIN

0 %

0 %

DRIVE

ENGINE

HYPERDRIVE

RAZOR WING

SUPERNOVA

DRIVE

COLOUR

10 %

100 %

COMP

COMP

SPEED

20 %

50 %

OUT

LEVEL

VELO

-0.1 dB

50 %

☐ KEY TRACK

WHAT IS IN HERE

03	QUICK START	Install it, hit it, pick a preset.
04	THE IDEA	Every kick lives in one space.
05	BODY	The drum itself: seven knobs.
06	HIT & ERA	The beater, the machine, the room.
07	TRANSIENT	More or less front, more or less tail.
08	DRIVE	Four engines, level-matched.
09	COMP & OUT	Punch, pump, level, touch and pitch.
10	THE DISPLAY & THE PAD	What the screen is telling you.
11	RECIPES	From one style to another.
12	PRESETS	Thirty-one, in three banks.
13	SPECIFICATIONS	And what the bench measured.

IN SHORT

Kickstart synthesises kick drums. All of them: the acoustic kit, the drum machines that made the records, and whatever techno, gabber and trap did next.

It does not do it with one simulation per machine. It does it with twenty-one controls that describe what every kick is made of — and then lets you walk from one drum to the next.

THE FIRST TEN SECONDS

Open it, click the big **HIT** pad, then press > beside the preset name until something makes you grin.

INSTALL, HIT, PLAY



The panel. The display and the HIT pad across the top; below, seven sections in the order the sound meets them: Body, Hit, Era, Transient, Drive, Comp, Out.

1 • INSTALL

Copy **Kickstart.vst3** into C:\Program Files\Common Files\VST3\ and rescan in your DAW. It files under Instrument / Drum. The standalone, **Kickstart.exe**, needs no installing.

2 • FIRST HIT

Click the **HIT** pad. The centre is the hardest hit, the rim softer. Space bar works too, and so does any MIDI note.

3 • PICK A PRESET

The < and > buttons step through the banks; the name opens the full list. Presets are level-matched, so you are comparing drums, not volumes.

4 • KEY TRACK

Off, every note plays the same kick — map it to whatever your drum rack sends. On, the note sets the pitch, with C1 at PITCH: load **Trap 808** and play a bass line.

5 • KEEP IT

SAVE and **LOAD** use Documents\Brokild patches\Kickstart.

ONE KICK, MANY DRUMS

An 808, a 909 and an acoustic kick are not three machines. They are three points in one space.

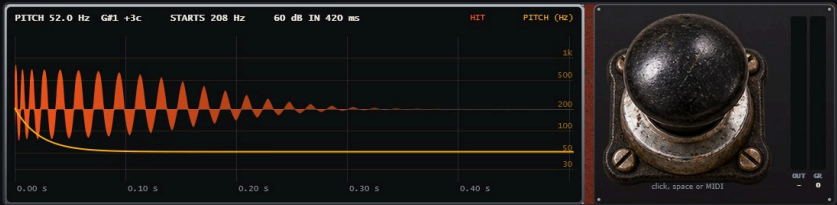
A kick is a drum whose pitch falls. That is nearly the whole story, and it can be told with a handful of physical axes:

AXIS	808	909	ACOUSTIC
How far the pitch falls · SWEEP	a little	two octaves	a little
How fast · BEND	slowly	in a flash	fast
How it rings · DECAY, CURVE	long and pure	shorter	short, damped
What it is made of · WAVE, SKIN	sine	triangle-ish	a real head
What hits it · CLICK, TONE	barely	a noise click	a beater
Where it is · GRIT, ROOM	dry	dry	a room

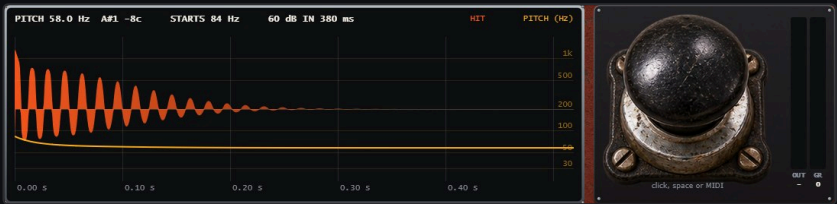
Move along any one of those and you walk from one drum to another. The recipes on page 11 are walks like that. Everything after — transient, drive, compression — is what a producer would do to the drum next.



808 Boom. A slow fall, a long tail.



909 Classic. Down two octaves in 22 ms.



Studio Kick. A real head, short and damped.

BODY

PITCH 25-160 Hz

Where the kick settles. The display shows the note name beside it, so you can tune the kick to the song.

SWEEP 0-48 semitones

How far above PITCH the hit starts. An 808 a little, a 909 a lot, gabber more still.

BEND 2-300 ms

How fast the pitch falls. Short is a snap, long is a zap.

DECAY 50 ms-4 s

The time to fall 60 dB. The display prints the result, so “60 dB in 420 ms” means exactly that.

THE FASTEST WAY TO A DIFFERENT DRUM

SWEEP and BEND together. Most of the difference between an 808 and a 909 is how far the pitch falls and how quickly. Move those two first, and everything else is detail.

CURVE 0-100 %

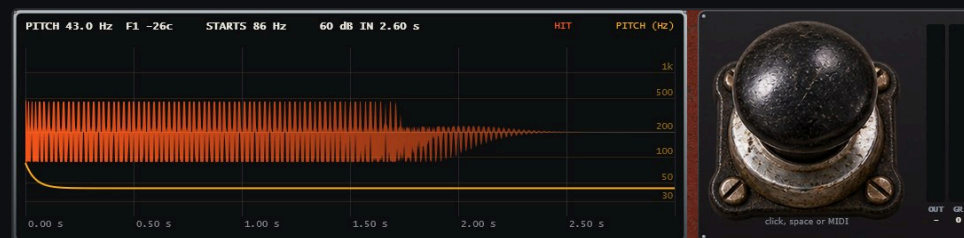
At 0 the kick dies the way a struck drum does, exponentially. At 100 it holds full level and then is gone — the modern sub.

WAVE sine → square

Sine, through triangle (the 909, around 35 %), to a rounded square. More harmonics, more presence on small speakers.

SKIN 0-100 %

From an electronic oscillator to a real drum head. It adds the circular membrane’s inharmonic modes and the tension glide: a real head starts sharp the harder you hit it, then settles.



CURVE at work. Trap 808 holds, then lets go.

HIT & ERA

HIT • THE BEATER

CLICK

The beater's level: how much of the moment of impact you hear on top of the drum.

tone

What the beater is. Low is a felt thud, then a wooden knock, then the 909's noise click, and at the top a digital tick.

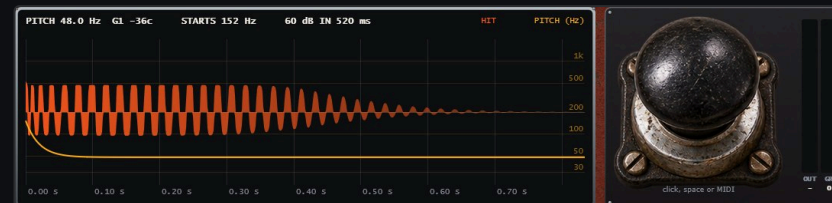
ERA • WHEN AND WHERE

GRIT

Lowers the sample rate and the bit depth. Around **30 %** sounds like the 12-bit machines — SP-12, Linn 9000, MPC60. Around **55 %** it is the 8-bit DMX and LinnDrum.

ROOM

The air around a real kit. Feed it into DRIVE and it stops being a room and becomes a techno rumble.



Techno Rumble. ROOM at 60 % into the drive, COLOUR pulled down to 35 %. The rumble is the reverb, distorted.

WHY ERA IS ITS OWN SECTION

The sample machines of the eighties did not sound the way they did because of their kicks. They sounded that way because of their converters. GRIT is the converter; the drum is still yours.

TRANSIENT

ATTACK bipolar

More or less of the first milliseconds. Up for a kick that cuts through a mix; down to round it off. Centre does nothing.

SUSTAIN bipolar

More or less of the tail. Down to tighten a boomy kick without touching DECAY; up to bring out the body and the room.

Transient shapers usually have to guess. They follow the audio with two envelope detectors and compare them, and on a low kick those detectors ride on the waveform itself — the gain wobbles at the pitch of the drum.

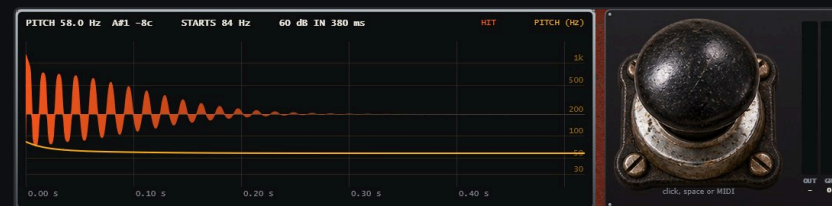
Kickstart does not have to guess. It is a synthesiser, so it **knows** its own envelopes, and the shaper reads those directly.

NO RIPPLE

Because nothing is detected from the audio, there is nothing to ripple. What is left over on a low kick measures **-79 dB**.

TRY THIS

Load **Psytrance Tick**: ATTACK is up and SUSTAIN is down, which is most of why it is all attack. Now swap them round.



Studio Kick. ATTACK +30 %, SUSTAIN -20 %: the beater forward, the ring pulled in.

DRIVE

Four engines, lifted from Brokild's own Battlestar Overdrive.

ENGINE	WHAT IT DOES
IDLE BURN	A warm, asymmetric tube. Weight and a second harmonic.
HYPERDRIVE	Harmonic injection. Adds harmonics above the sub, so a kick that lives at 45 Hz still reads on a phone speaker.
RAZOR WING	Tanh into a hard clip. Flat-topped and mean — this is gabber.
SUPERNOVA	Fold, crush and chaos. Broken on purpose.

DRIVE

How hard the engine is pushed.

COLOUR

The drive's low-pass. Fully open is no filter at all; pull it down for a darker, rounder distortion.



Gabber. RAZOR WING at 90 %: the sine has become a square.

LEVEL-MATCHED

Every engine is matched by measurement, so turning DRIVE up changes the character, not the loudness — within 2.4 dB across every engine and every setting. Judge it by ear, not by volume.

COMP & OUT

COMP

COMP

Threshold and ratio on one knob, with its own make-up gain, so turning it up makes the kick denser, not quieter.

SPEED

Slow lets the hit punch through before the compressor grabs the body. Fast flattens the whole thing and pumps. The compressor looks 1.5 ms ahead, so even fast it catches the front.

RETRIGGER

Hit it again while it is still ringing and the old kick is choked over 3 ms, the way a real voice would be, rather than clicking.

OUT

LEVEL

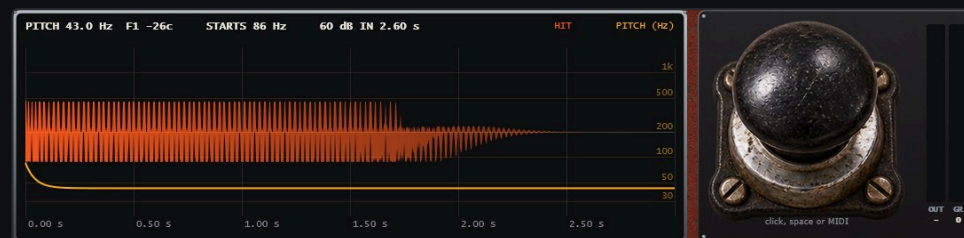
Output level. A soft ceiling holds everything under 0 dBFS however hard the section before it is pushed.

VELO

How much a hard hit differs from a soft one: louder, brighter, and bending further.

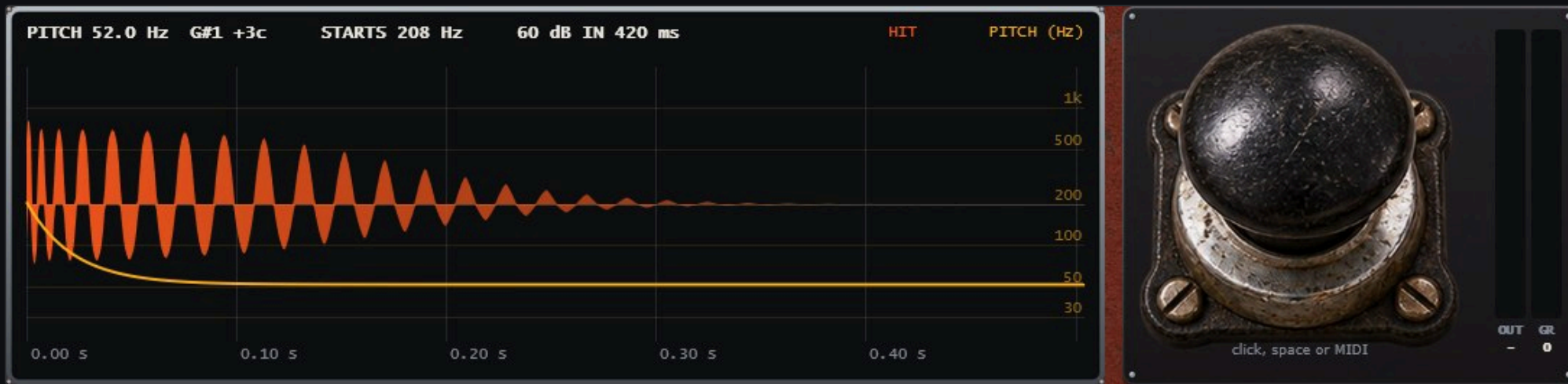
KEY TRACK

Off: every note plays the same kick. On: the MIDI note sets the pitch, with **C1** = **PITCH**, so the kick plays in tune as an 808 bass line.



Trap 808 ships with KEY TRACK on.

THE DISPLAY & THE PAD



909 Classic. The hit in orange, its pitch in amber on a logarithmic scale; the readouts along the top.

THE HIT

The whole kick, drawn in orange, redrawn every time a control moves. You see a change before you hear it.

THE PITCH

The amber curve is the pitch falling to PITCH. The top line reads where it settles, the note, where it starts and the decay time.

PAD & METERS

Click the pad: centre hardest, rim softer. Space and MIDI work too. OUT is the output level; GR is the compressor at work.

RECIPES

From one style to another, a few knobs at a time.

808 → 909

1. Start on **808 Boom**.
2. Raise SWEEP towards 24 semitones and shorten BEND to about 20 ms.
3. WAVE to 35 %: the triangle.
4. CLICK up, TONE towards the noise click. Shorten DECAY to taste.

TECHNO RUMBLE

1. A 909-ish kick with a longer DECAY.
2. ROOM up to about 60 %.
3. DRIVE up, IDLE BURN: the room goes through the drive.
4. COLOUR down until the tail rumbles instead of hisses. Some COMP.

ANY KICK → A PLAYABLE BASS

1. CURVE up past 50 %, DECAY to two seconds or more.
2. HYPERDRIVE with a little DRIVE, so it reads on a phone.
3. KEY TRACK on. C1 plays at PITCH.
4. Keep SWEEP modest, or every note starts with a zap.

ELECTRONIC → ACOUSTIC

1. SKIN up past 50 %.
2. SWEEP down to a few semitones.
3. TONE down, towards felt or wood.
4. A little ROOM. Then SUSTAIN down if it rings too long.

GABBER

1. SWEEP up high, three octaves.
2. CURVE up, so the kick holds instead of dying.
3. WAVE up, then RAZOR WING with DRIVE near the top.
4. CLICK and TONE up. Play it at 180.

CLEAN → 1988

1. GRIT to about 30 % for 12-bit, 55 % for 8-bit.
2. A touch of IDLE BURN, COLOUR down a little.
3. COMP up with a slow SPEED, so the hit punches through.
4. Put it under a sampled break.

PRESETS

ACOUSTIC

Studio Kick

A mic'd 22-inch kick: beater, head and a little room.

Jazz Kick

18 inches, resonant head on, felt beater, lots of tone.

Rock Kick

26 inches, hard beater, pushed.

Felt Beater

Soft and round.

Towel In The Drum

Seventies dead: all thud.

Kit In A Room

Heard from the other side of the room.

Init

A plain electronic kick to start from.

CLASSIC

808 Boom · 808 Short

The long, low, almost pure boom — and with the decay down.

909 Classic · 909 Hard

The sweep, the triangle, the click — and pushed through the desk.

707 / 727

Brittle mid-eighties sample machines.

LinnDrum · DMX

A real kick, 8-bit — and bigger, airier.

Linn 9000

12-bit and cleaner.

SP-12 Boom Bap · MPC60 Thump

12-bit, pushed and compressed.

KR-55 · CR-78

Short, woody analogue; the soft preset-rhythm knock.

Drum Computer

Movement Systems: analogue meets digital.

MODERN

Techno Rumble · Hard Techno

The reverb, distorted; clipped with a room.

Gabber · Hardstyle

A sweep held flat and clipped; the punch, then the tail.

Dubstep Punch

A bright click on a short clean sub.

Trap 808

Key tracking on: play it as a bass.

Psytrance Tick

Short, tight, all attack.

Industrial Crush

SUPERNOVA and grit.

House Thump · Lo-Fi Dust

Four to the floor; left in a drawer since 1991.

Clean Sub

A long held sub that still reads on a phone.

The classic bank follows MusicRadar's ten drum machines that changed music history. Each preset is in the spirit of its machine.

SPECIFICATIONS

Format	Windows VST3, 64-bit, and a standalone application
Type	Instrument / Drum. MIDI in, stereo out
Sound	Fully synthesised. No samples
Controls	21, in seven sections, plus KEY TRACK
Drive engines	IDLE BURN, HYPERDRIVE, RAZOR WING, SUPERNOVA
Presets	31 factory in three banks, plus Init
User patches	Documents\Brokild patches\Kickstart
Sample rates	Any. Bench-tested at 44.1, 48 and 96 kHz
Licence	AGPLv3. Source at github.com/peterboggild/BrokildApps
Build	260926.1

MEASURED

Tuning against PITCH	within 0.04 cents
Oversampling	4x, always
Aliasing, brightest setting	-90 dB
DECAY against its -60 dB time	within 10 %
Transient shaper ripple	-79 dB
Drive level change	within 2.4 dB
Retrigger choke	3 ms
Latency at 48 kHz	95 samples, 2 ms
CPU, everything on	about 4 % of a core

Latency is reported to the host and compensated. Presets are level-matched to -10 dB RMS over the first 150 ms.

CREDITS

NOW GO AND HIT SOMETHING

This VST comes from the curved mind of Professor Brokild, and was programmed with VScode, ChatGPT and Claude. Use of this is on your own peril. It may generate unique, beautiful, scary or useless sounds for you, and it may crash your DAW project. Have fun, play unsafely, unwisely, and unboringly.

Cheers, Peter Bøggild, September 2026.

Drum machine names are trademarks of their owners, used only to describe the sound a preset is in the spirit of. Kickstart contains no samples of them.

BROKILD

Free, and free to pass on.
peterboggild.github.io/BrokildApps

BUILD 260926.1 • WINDOWS VST3

